

BlastSmart App Design

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ABSTRACT

The mining industry gradually relies on digital technologies to improve efficiency and safety of blasting operations. This study presents preliminary research findings on two core functionalities of the BlastSmartApp: the surface blasting calculator and the fragmentation calculator. The main objective of this research is to develop clear algorithms, flowcharts, and pseudocodes for both calculators to support their implementation in the app. The surface blasting design calculator uses blasting diameter, average in-hole density, spacing to burden ratio, and bench height as inputs to calculate burden, stemming length, and actual powder factor. The fragmentation calculator uses average in-hole density, bench height, relative effective energy of the explosive, and the stemming length to estimate rock fragmentation size based on established blast fragmentation models. The study found that the surface blasting design calculator can be developed using standard blast design equations, while the fragmentation calculator can be implemented using the widely accepted Kuz-Ram fragmentation model to predict rock fragmentation. The identified algorithms provide a suitable foundation for developing the BlastSmartApp, with flowcharts and pseudocode supporting the accurate implementation of both calculators.

BACKGROUND

Blasting is an essential mining operation that requires accurate calculations to ensure efficient rock fragmentation, safety, and cost-effective production. Traditionally, blast designs have been performed using manual calculations and paper-based methods which can be time-consuming and prone to errors. To improve efficiency Dwala Global Resources proposes the development of the BlastSmart App a mobile application that assists blasting engineers with blast design calculations and parameter validation. The app will automate calculations predict blast outcomes and support decision-making for both surface and underground mining operations. By digitizing

blast design processes the application aims to improve accuracy, productivity, and safety in the mining industry.

LITERATURE REVIEW

Surface blast design involves selecting appropriate blasting parameters, such as burden, stemming length, and powder factor, to achieve safe, efficient, and cost-effective rock breakage while minimizing adverse effects such as flyrock and ground vibration (Dinis da Gama, 1990). Accurate blast design improves blasting performance and overall mining efficiency. Rock fragmentation is an important measure of blast performance because it affects loading, hauling, and crushing operations. The Kuz-Ram model is widely used to predict the mean fragmentation size from blast design parameters (Cunningham, 1983). Therefore, the Surface Blast Design Calculator and Fragmentation Calculator developed in this project automate key blast calculations, reduce manual computation, and assist engineers in planning efficient blasting operations.

DESIGN

The design of the BlastSmart application was guided by a structured approach that separates the system into two independent modules: the Surface Blast Design Calculator and the Fragmentation Calculator. This modular design simplifies the development process by allowing each calculator to perform a specific task while maintaining the overall functionality of the application. The system was designed to accept user inputs, process the data using predefined mathematical relationships, and present the calculated outputs in an organised manner. This approach improves the readability, maintainability, and scalability of the application. To ensure the correctness

of the system before implementation, algorithms and pseudocodes were developed for both calculators. The algorithms describe the logical sequence of operations required to transform the user inputs into the desired outputs, while the pseudocodes provides a structured representation of this logic independent of

any programming language. Developing these design components before coding made it easier to identify the required calculations, organise the program flow, and minimise logical errors during implementation.

MATHEMATICAL EQUATIONS

The mathematical equations used in the application were derived from established surface blasting design principles and the Kuz-Ram fragmentation model. These equations form the computational core of the application and are used to calculate the required blast design parameters and the expected mean fragmentation size from the user-provided inputs.

mathematical formulas that were used:

Formulas used:

i) Surface Blast Design Calculator formulas:

1. *Stemming length* = $(25 \times \text{blasting diameter}) + 1000$
2. *Burden* = $\sqrt{\frac{\text{linear charge mass}(MC) \times \text{charge length}(CL)}{\text{spacing to burden ratio}(s) \times \text{Bench Height}(BH) \times \text{Technical powder factor}(PFtech)}}$
 - $MC = \frac{P \times BD^3}{1273}$
 - $CL = BH - \text{Stemming length}$ (assuming there is no sub-drill)
 - $PFtech = 0.3$ (standard $PFtech$ for limestone)
3. *Actual powder factor* = $\frac{BH \times \text{Stemming length} \times MC}{\text{Burden} \times \text{Spacing}(S) \times \text{Hole Length}(H)}$
 - $\text{spacing} = a \times \text{Burden}$
 - $\text{Hole Length} = \text{Charge Length} + \text{Stemming Length}$

ii) Fragmentation calculator:

$$1. \text{mean fragment size} = A(K)^{-0.8} Q^{0.167} \left(\frac{115}{E}\right)^{0.633}$$

Figure 1:

RESULTS

Process The surface blasting design calculator was developed by processing the inputs in parameters (BD, P Sratio, and BH) using standard blast design equations to calculate burden, stemming length, and the actual powder factor. The fragmentation calculator processes the input parameters (p, BH, REE, and SL) using the KUZ-RAM model to estimate the mean fragment size. Both processes were represented

Inputs	Outputs
Blasting diameter (BD)	Stemming length
Average in-hole density (p)	Burden
Spacing to burden ratio (S/B ratio)	Actual powder factor
Bench height (BH)	

Figure 1.1 : Surface Blasting Design calculator

Inputs	Outputs
Average in-hole density (p)	Mean fragment size
Bench height (BH)	
Relative effective energy of explosive (E)	
Stemming length (SL)	

Figure 1.2 : Fragmentation Calculator

Figure 2:

using algorithms, flowcharts and pseudocodes for implementation in the BlastSmartApp.

DISCUSSION OF RESULTS

The results show that the Surface Blasting Design Calculator successfully identifies the key input parameters required for blast design, namely blasting diameter, average in-hole explosive density, spacing-to-burden ratio, and bench height. Using these inputs, the calculator determines burden, stemming length, and actual powder factor, which are critical parameters for achieving effective rock fragmentation and controlling blasting costs. The results indicate that standard blast design equations can be incorporated into the BlastSmartApp to automate calculations that are traditionally performed manually. The Fragmentation Calculator uses explosive density,

bench height, relative effective energy and stemming length to estimate mean fragment size through the Kuz-Ram fragmentation model. This result demonstrates that the calculator can predict blast fragmentation outcomes before blasting takes place allowing engineers to optimize blast designs and improve downstream processes such as loading, hauling, and crushing. The use of the Kuz-Ram model provides a reliable and widely accepted method for estimating fragmentation performance.

CONCLUSION

This project successfully developed two engineering calculators that automate common blasting. The

application reduces computational time, improves calculation accuracy and provides easy-to-use interface for mining engineers. The project demonstrates how computer programming can support engineering decision-making while reducing manual errors during blast design.

FUTURE WORK

Although the development application successfully performs the required surface blasting design and fragmentation calculations, several improvements can be made in future versions to increase its functionality and usability. a graphical user interface can be developed to provide a more user-friendly experience compared to the current command line interface. Additional explosive types such as ANFO, emulsion explosives and TNT, can be incorporated with their respective relative efficient energy values to allow users to select the appropriate explosives for different mining conditions. the application can also be enhanced by integrating advanced fragmentation prediction models, such as the Kum-Ram model to improve the accuracy of mean fragment size estimation. The program can also be extended to include additional blasting design parameters, including sub-drilling, rock density, rock factor and drilling patterns , thereby providing more comprehensive blast design calculations.

Bibliography

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